DPCN · ASSIGNMENT 1

Opinion Network Formation

A similarity network built from a class survey on Technology, Education, Ethics, and Environment

Team name	Deadlines Passing, Care Not
Members	Aviral, Varun, Arihant
GitHub	https://github.com/avi1o1/DPCN_A1
Deliverables	A reproducible pipeline, five opinion networks, static figures, and a self-contained interactive dashboard.

Overview

Each of the 87 respondents is a node. We connect two people when their survey answers are very similar. We then do three sets of analysis on this network. First, we measure the structure. This covers degree, clustering, components, centrality, and communities. Second, we run two dynamic processes from the course. These are the DeGroot opinion average and an SIS contagion. Third, we run two null models. These test whether the structure is more than chance. This report defines each concept in a shaded box before it uses the concept.

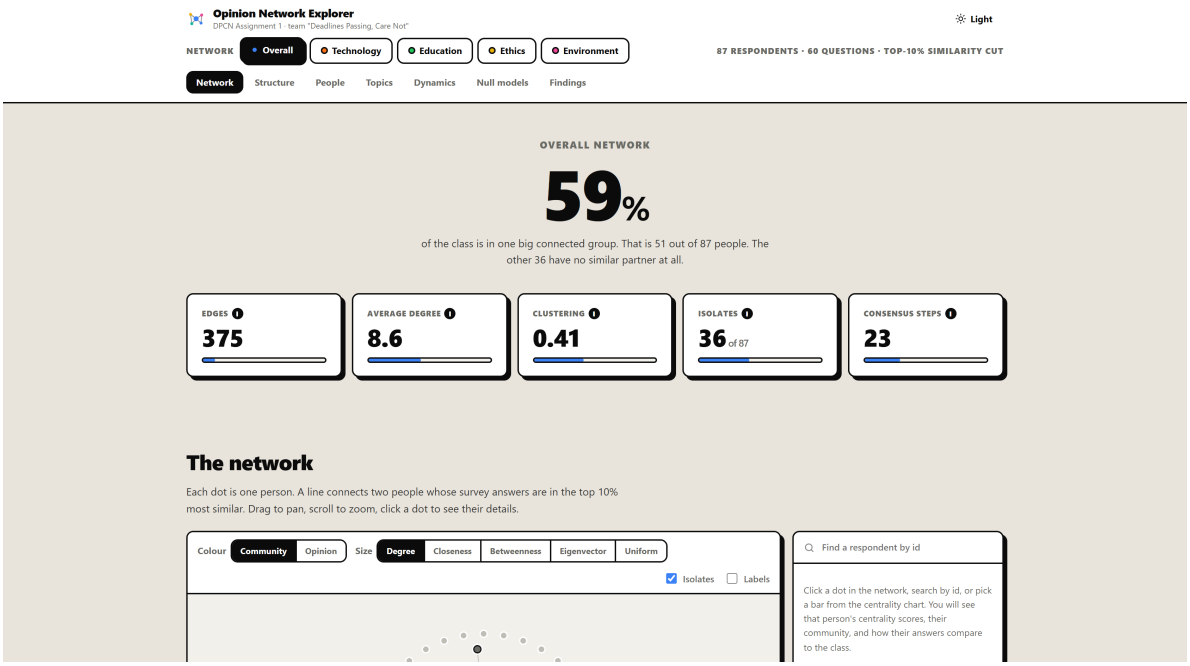


Figure 1. The interactive dashboard. It opens in any browser and needs no server. The network picker changes every number, chart, and diagram to the chosen network. The main result is that 59% of the class (51 of 87) is in one connected group. The other 36 have no strong-similarity partner.

1 Dataset Documentation

The raw survey. The survey has 97 responses to 60 opinion statements. There are 15 statements in each of four topics. A short code marks the topic of each statement. **T** is Technology, **E** is Education, **S** is Ethics, and **V** is Environment. Each statement uses a five-point Likert scale from *Strongly Disagree* to *Strongly Agree*. The survey also gives a "No Comments" option.

Preparation.

- *Likert to number.* We map the scale to a symmetric integer. Strongly Disagree is -2. Disagree is -1. Neutral is 0. Agree is $+1$. Strongly Agree is $+2$. The symmetry makes 0 a true neutral. A mean score then shows a direction and a strength at the same time.
- *"No Comments" is missing, not neutral.* We record it as a blank, not as 0. A refusal to answer is not the same as "I am neutral." There are 62 blanks out of $87 \times 60 = 5220$ (1.19%). They affect 19 respondents.
- *Quality filter.* We remove any respondent who answered fewer than half of the 60 questions. This removes 10 sparse responses. It leaves **87 respondents**. We use these 87 everywhere below.
- *Result.* The cleaned data is an 87×60 table of scores in $[-2, +2]$, with a blank for each "No Comments." Each question keeps its topic and its text.

Missing values. Two rules are correct, and we use both on purpose. The structural work must agree with the graph. For this work, we replace a blank with 0 before we average. This is the same step that the edge builder uses, so the opinions and the edges use the same numbers. The descriptive statistics instead drop the blank from a person's average. The two rules differ by at most 0.01 on the class mean. We never compare a number from one rule with a number from the other.

2 Pipeline Followed

CONCEPT · opinion vector and cosine similarity Each respondent becomes a **vector** of up to 60 numeric answers, $\mathbf{x}_i \in [-2, 2]^{60}$. Blank cells become 0. The similarity of two people is the **cosine** of the angle between their vectors.

$$s_{ij} = \cos(\mathbf{x}_i, \mathbf{x}_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \, \|\mathbf{x}_j\|} \in [-1, 1].$$

The value is $+1$ when two answer patterns point the same way. It is 0 when they are unrelated. It is -1 when they are opposite. The cosine rewards a shared pattern across all 60 questions. It does not reward only a shared average level.

From answers to a network. A node is a respondent. An edge means that two people hold similar opinions. We compute the full 87×87 cosine-similarity matrix. We then keep only the **top 10% most similar pairs** as edges. This is the 90th-percentile cut. A hard threshold is necessary. Almost everyone in this class agrees on average. A loose threshold gives a near-complete graph with no information. The 90th-percentile rule gives exactly **375 edges**. We build **five networks** with the same rule. One network is *overall* and uses all 60 questions. Four networks use one topic each with 15 questions. The same rule means that any difference between the networks comes from the opinions. A sensitivity sweep in §3.5 tests the threshold.

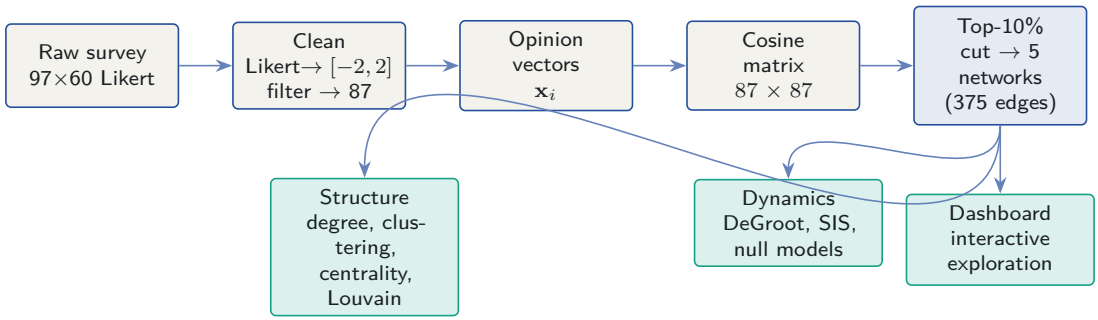


Figure 2. The pipeline. We clean the survey, build the opinion vectors, compute the cosine matrix, and cut the top 10% of pairs into five networks. The structure, the dynamics, and the dashboard all read these networks. No later stage rebuilds the graph.

What runs on top. One stage computes all structural metrics. A second stage computes the dynamics. A verification stage re-checks the maths with 21 independent tests. A final stage assembles the dashboard. The later stages read the saved networks and never rebuild the graph.

3 Analysis and Visualizations

3.1 Structure of the overall network

> **CONCEPT · degree, density, clustering, components** The **degree** k of a node is its number of edges. The **average degree** $\langle k \rangle$ is the mean degree of the class. The **density** is the fraction of all possible edges that are present. The **clustering coefficient** measures how often two neighbours of a node also connect to each other. Its range is 0 to 1. A **connected component** is a set of nodes that can all reach each other. The largest one is the **giant component**. Inside it, the **diameter** is the longest shortest path. The **average path length** is the mean shortest path. It shows how many steps separate two connected people.

Table 1 gives these metrics for all five networks. The overall network is sparse, with a density of 0.10. It is dense where it connects, with a clustering of 0.41. The connected people form tight triangles. 36 of the 87 respondents are **isolates**. An isolate has no strong-similarity partner. The giant component is compact. Two connected people are only about 1.9 steps apart.

Metric	Overall	Technology	Education	Ethics	Environment
Nodes / Edges	87 / 375	87 / 375	87 / 375	87 / 375	87 / 375
Density	0.100	0.100	0.100	0.100	0.100
Avg. degree $\langle k \rangle$	8.62	8.62	8.62	8.62	8.62
Avg. clustering	**0.408**	0.387	0.396	0.363	0.375
Components	37	21	23	42	43
Giant component	58.6%	**74.7%**	72.4%	51.7%	50.6%
Giant diameter	4	5	6	4	5
Giant avg. path	1.88	2.20	2.24	1.78	1.93
Modularity Q	0.108	0.242	0.263	0.151	0.079

Table 1. Structural metrics per network. All five networks share the node count, the edge count, and the density. So every difference is an opinion effect.

3.2 Central people, and how the network splits

> **CONCEPT · four centralities** **Degree** centrality counts direct connections. **Closeness** centrality measures how short the average path is from a node to all others. **Betweenness** centrality counts how often a node lies on the shortest path between two others. It shows the value of a node as a bridge. **Eigenvector** centrality gives more weight to a node that connects to other important nodes. Degree, closeness, and eigenvector reward a position in the core. Betweenness rewards a position between groups.

Degree, closeness, and eigenvector centrality pick almost the same five people (114, 67, 94, 105, 110). All five score above the class mean on every topic. They are the uniform strong-agreement answerers. Their 60-value vectors are similar to many others. Betweenness centrality gives a different answer. Its top node is 41. Respondent 41 has a maximum Technology score but only an average Education score. This mixed profile makes 41 a bridge between sub-groups (Figure 3, right). The most connected person and the best bridge are not the same person. One centrality is not enough.

> **CONCEPT · community, modularity, Louvain** A **community** is a group of nodes with many edges inside and few edges to the outside. **Modularity** Q scores a given grouping. It measures how many more edges fall inside the groups than a random graph with the same degrees gives. Its

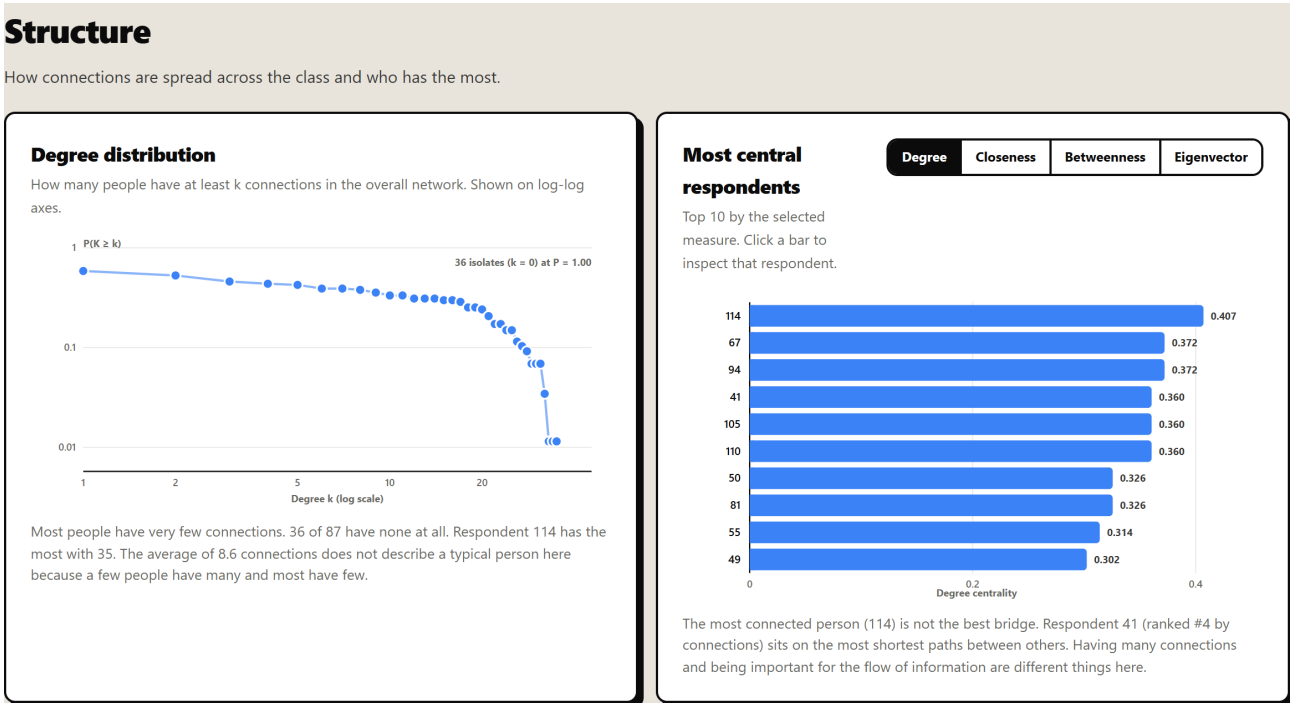
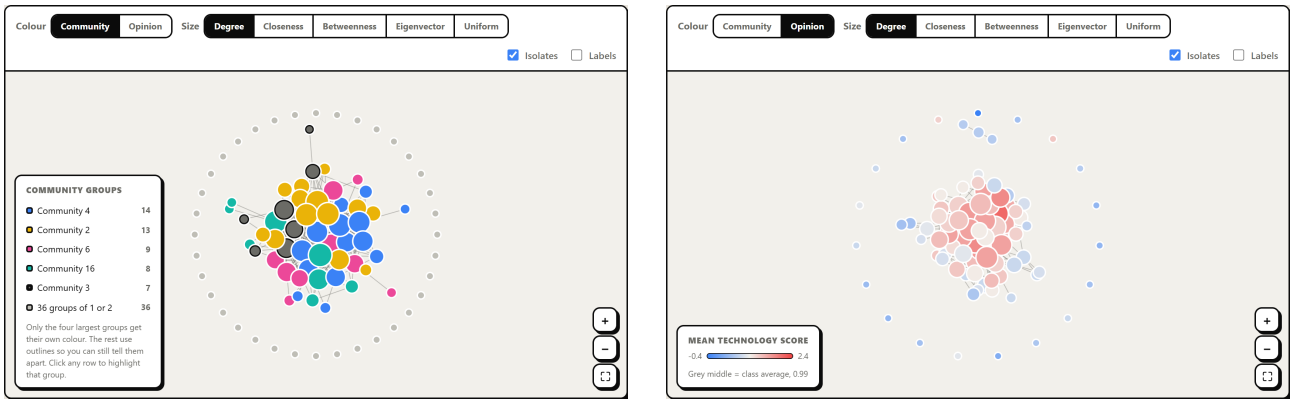


Figure 3. *Left:* the degree distribution on log-log axes as a complementary CDF $P(K \geq k)$. It has a heavy tail. A few hubs (respondent 114 has $k = 35$) sit above a long tail of low-degree nodes. The 36 isolates sit at $k = 0$, $P = 1$. A log axis cannot draw them, so the chart states this. The mean of 8.6 fits almost nobody. *Right:* the ten most central respondents by degree.

> range is $-\frac{1}{2}$ to 1. The **Louvain** algorithm looks for the grouping with the highest Q. A higher Q means clearer groups.



(a) Overall, coloured by community. **(b)** Technology, coloured by opinion.

Figure 4. The node-link view. Node size scales with degree. The giant component sits in the centre. Small components ring it. Isolates sit on the outer rim. **(a)** The four largest Louvain communities have colours. The 36 singletons and pairs are grey on the rim. This gives the dense core with an isolate halo. **(b)** Colour by Technology score shows the main pattern. The strong-agreement answerers (red) form the centre. The lukewarm answerers (blue) sit on the edge.

Louvain finds many small communities in the overall network. The modularity is low ($Q = 0.108$). These communities show different levels of agreement. They are not opposite camps. Every community has a positive mean on every topic (Figure 5). Only one group (C6) falls below the positive class mean, and only on Technology. This group is mildly cool on technology. It is not against technology.

3.3 Cohesion tracks opinion *variance*, not opinion strength

The per-topic numbers go against the raw averages. Ethics (+1.28) and Environment (+1.37) have the strongest average agreement. But they have the most broken networks, with a giant component near

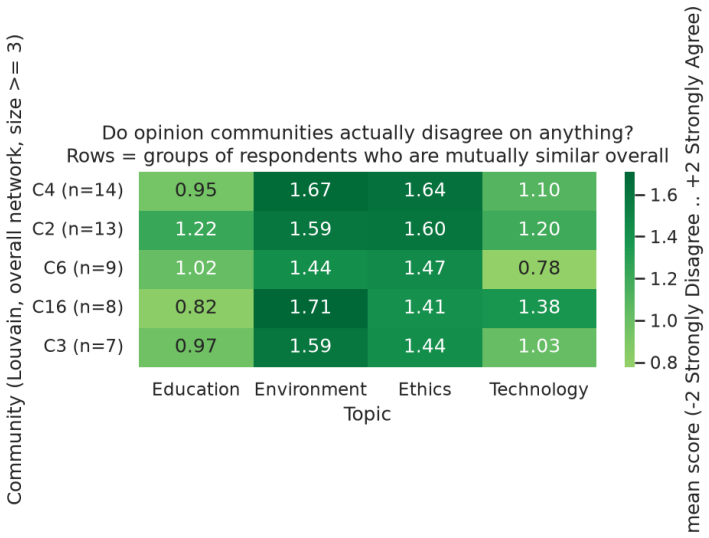


Figure 5. Mean score of each large overall-network community, per topic. Every cell is positive. The communities raise or lower agreement. They do not disagree. A permutation test in §4 shows that these topic means do not separate the communities.

52% and 51%. Technology (+0.99) and Education (+0.91) agree less on average. But they stay the best connected, at 75% and 72%. The cause is the spread, not the mean. Cosine similarity depends on the full pattern across the 15 questions of a topic. Technology and Education opinions are tightly grouped, with a standard deviation of 0.37 and 0.35. So most vectors are similar and the graph stays connected. Ethics and Environment opinions are strong but more spread out, with a standard deviation of 0.45 and 0.47. So the same cut breaks the class into fragments. The statement "the class agrees more on Environment than on Technology" holds for the raw average. It fails for the network.

3.4 Dynamics on the network

> **CONCEPT · DeGroot opinion average and stability** Give each respondent a start opinion, which is their mean score. At each step, each respondent changes their opinion to the mean of the opinions of their neighbours. All respondents do this at the same time. The rule is $\mathbf{x}(t+1) = D^{-1}A\,\mathbf{x}(t)$, where A is the adjacency matrix and D is the degree matrix. This is the **DeGroot model**. The **eigenvalues** of the update matrix control the result. The course rule says that a linear map is stable when the relevant eigenvalue has a magnitude below 1. The **SLEM** (second-largest eigenvalue modulus) sets the speed. The **algebraic connectivity** $\lambda_2 = 1 - \mu_2$ measures how well a component connects. A larger λ_2 gives a faster consensus.

The overall network has 37 components. So there is no single class consensus. The connected core of 51 people reaches **1.409** in **23 steps**. The 36 isolates never move, because an isolate has no neighbour to average with. The convergence speed follows the algebraic connectivity exactly for all five networks (Table 2). This confirms the course stability rule on real data. The edge weights have almost no effect. The weighted consensus is 1.4099. The unweighted consensus is 1.4085. Row normalisation removes the small 0.85 to 0.95 range of the kept similarities.

Network	λ_2	SLEM	Steps	Core size	Consensus
Overall	0.627	0.397	23	51	1.409
Ethics	0.455	0.545	31	45	1.350
Technology	0.318	0.682	53	65	1.313
Education	0.264	0.736	61	63	1.275
Environment	0.229	0.771	77	44	1.349

Table 2. DeGroot consensus. Every SLEM is below 1, so every map is stable. The order of λ_2 matches the order of the measured steps with no exception.

CONCEPT · SIS epidemic threshold Model a strong opinion as a contagion. A person is *infected* or *susceptible*. An infected person infects each neighbour at rate β. An infected person recovers at rate γ. The course result gives a critical ratio $(\beta/\gamma)_c = 1/\lambda_{\max}$, where $\lambda_{\max}$ is the largest eigenvalue of the adjacency matrix. A denser core gives a larger $\lambda_{\max}$. A larger $\lambda_{\max}$ gives a lower threshold. A lower threshold means that an opinion spreads more easily.

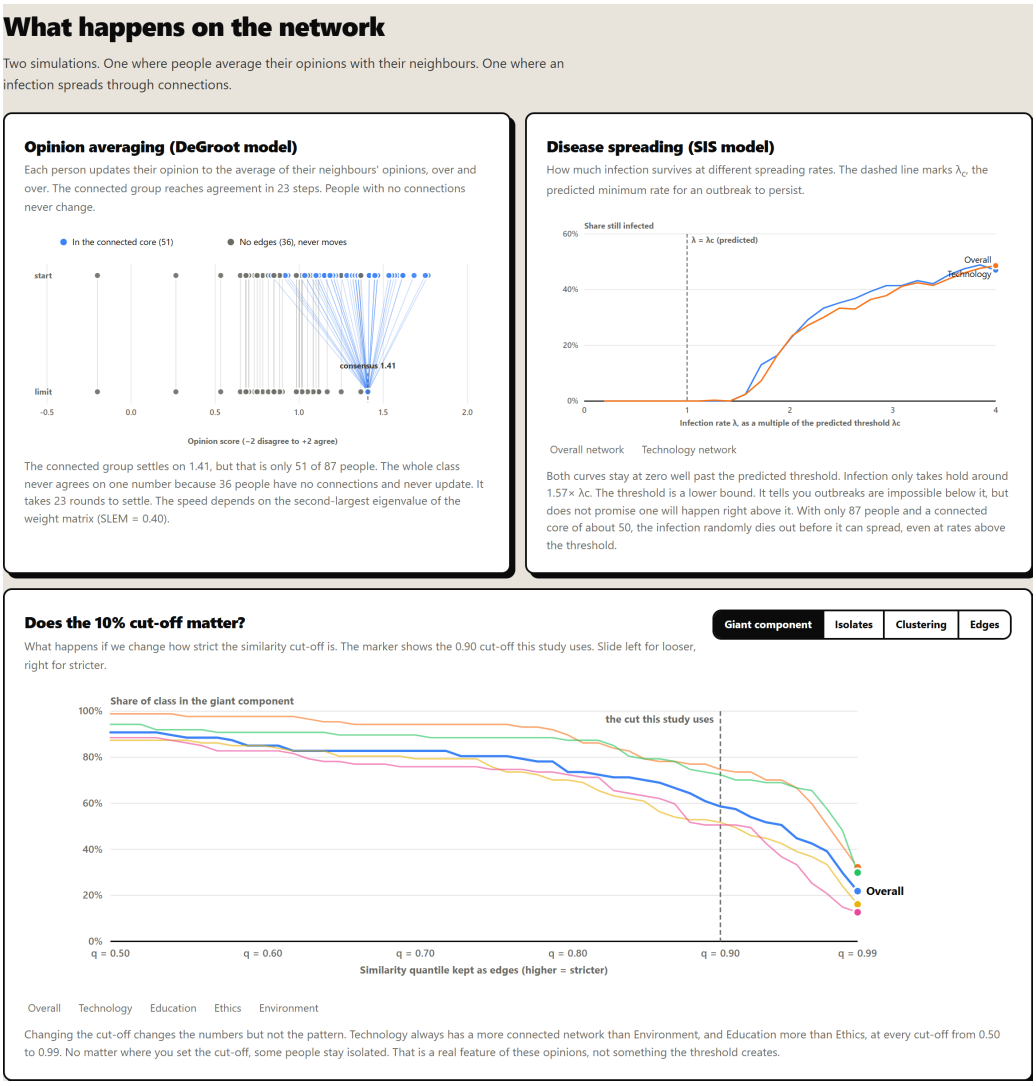


Figure 6. Dynamics. *Left:* DeGroot. The connected core funnels onto 1.41. The isolate lines stay flat. *Right:* the simulated SIS transition starts at about 1.6× the predicted bound λ_c. The bound is a lower bound, as theory requires. *Bottom:* the threshold sensitivity sweep. The topic connectivity order holds at every cut.

3.5 Does the 10% cut-off matter?

We apply the same rule at every quantile from 0.50 to 0.99 (Figure 6, bottom). Two results hold. First, the topic order is stable at every cut. Technology stays above Environment. Education stays above Ethics. This order is a property of the data, not of the 90th percentile. Second, the isolates are only partly an artifact. Their count falls from 36 at q=0.90 to 8 at q=0.50. It never reaches zero. Eight people stay below the median similarity to everyone. A looser cut also raises the clustering to 0.75 and makes the graph half complete. On such a graph, community and centrality lose their meaning. So q=0.90 is a tested compromise, not an assumption.

4 Results and Discussion

What the network shows about class opinions. This class is agreeable. The mean opinion is positive on all four topics. The raw data shows no opposite camps. The network shows structure that

the averages hide.

1. **No global consensus, and the drop-outs are the moderates.** Under DeGroot the 51-person core agrees at 1.409. The 36 isolates never move. The core started at a mean of 1.309. The isolates started at 0.872. So the average process removes the more moderate half of the class. This is a structural echo-chamber effect. The averaging model does not cause it.

2. **Cohesion is a variance story.** Section 3 shows that Technology and Education look more connected than Ethics and Environment. This holds although their scores are lower. The cause is the smaller spread. Raw agreement is not the same as social cohesion.

3. **Spread is easiest where the class is most broken.** Environment and Ethics have the lowest epidemic thresholds, at $1/\lambda_{\max} = 0.042$ and 0.044. But they are the least connected topics (Table 3). The edge count is fixed at 375. So a smaller giant component packs almost the same edges into fewer people. This raises $\lambda_{\max}$. Fragmentation and easy spread are not opposites here. This is a real property. It is also a warning about the fixed-edge method.

Network	$\lambda_{\max}$	$1/\lambda_{\max}$	Giant size	Giant density
Environment	23.92	0.0418	44	0.395
Ethics	22.45	0.0445	45	0.378
Overall	22.06	0.0453	51	0.294
Technology	18.17	0.0550	65	0.179
Education	17.74	0.0564	63	0.191

Table 3. Epidemic thresholds. The order is the reverse of the connectivity order in Table 1. The most broken topic spreads most easily.

> **CONCEPT · null models and the z-score** A null model answers one question. Does a random network with no opinion structure give the same value? The **Erdős–Rényi** (ER) model keeps only the number of edges. It places them at random. The **configuration model** is stricter. It keeps the exact degree of each person. It only changes who connects to whom. The **z-score** is $z = (\text{real} - \text{null mean})/\text{null s.d.}$ It shows how far the real value is from the random values. A value of $|z| < 2$ means that the real value looks random.

The structure is in the degree sequence (Figure 7). Against 500 ER graphs the network is unusual on every measure. The clustering z-score is +27 and the giant-fraction z-score is −362. Against 500 degree-preserving rewirings the network is ordinary. The clustering, the giant component, and the modularity all fall inside $|z| < 2$. So once you fix the number of edges of each person, chance gives the rest of the structure. For this construction, this is the same statement as "the connected core is the group that agrees with everything." The level of agreement sets the degree. After you fix the degrees, nothing is left to explain. This does not weaken the structural analysis. It gives the reason for it.

> **CONCEPT · permutation test and p-value** This test asks whether the communities are more similar in opinion than chance gives. It shuffles the community labels 10,000 times. It keeps the group sizes fixed. Each time it measures the opinion variance inside the groups. The **p-value** is the fraction of shuffles that beat the real grouping. A small p-value (below 0.05) means that the real communities are tighter than random.

The communities are real, but only at the fine grain. On the 60 question scores the Louvain grouping is much tighter than chance ($z = -6.57$, $p < 0.0001$). On the 4 topic means it is not different from random ($z = -1.26$, $p = 0.108$). Louvain groups people by their detailed answer pattern. It does not group them by their broad position on the four topics. So the community-by-topic heatmap (Figure 5) describes each group. But the topic means do not define the groups. We do not present them as definitions.

Summary. This class is agreeable across all four topics with no real factions. The network adds four corrections to the simple reading. First, low variance drives the apparent cohesion, so Technology and

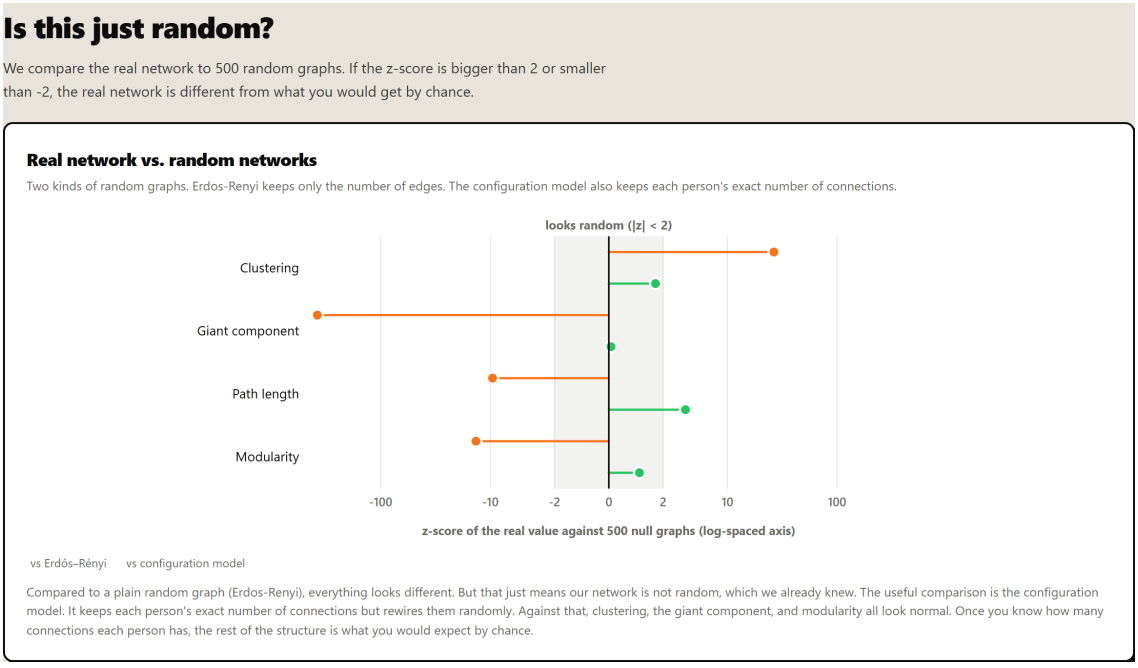


Figure 7. The real network against 500 random graphs of each type. The z-axis is symmetric-log. The shaded band is the $|z| < 2$ range that looks random. Erdős–Rényi (orange) fails every measure. The configuration model (green) matches clustering, the giant component, and modularity. The structure is in the degrees.

Education out-connect Ethics and Environment even with lower means. Second, the connected core is a behaviour type. It is the group with uniform strong agreement. It is not a demographic or a topic camp. Third, the centrality measures disagree about who matters, and that disagreement is itself a result. Fourth, Ethics and Environment opinions move together (respondent-level $r = 0.70$) and form one values axis. Technology and Education are almost independent ($r = 0.29$).

5 Individual Contribution

The course grades this assignment individually. The commit history shows the split below.

Member	Tasks completed
Aviral	Cleaned and interpreted the survey data. Built the networks with cosine similarity and the top-10% cut, for all five networks. Ran the structural analysis: degree, clustering, components, the four centralities, and the Louvain communities. Made the static figures. Found the topic-variance result and the centrality result.
Varun	Ran the dynamics and the mathematical analysis. Ran the DeGroot average and the spectral stability analysis. Computed the SIS epidemic threshold and the simulation. Built the Erdős–Rényi and configuration null models. Ran the community permutation test. Ran the edge-threshold sensitivity sweep. Wrote the verification checks.
Arihant	Built the self-contained interactive dashboard: hand-drawn charts, a node-link view, all five networks, community and opinion colour, an isolate toggle, a respondent lookup, and the dynamics and null-model panels. Exported the dashboard data. Wrote this report and produced its figures.

Table 4. Individual contributions.

Reproducibility. Everything in this report comes from the linked repository. The pipeline runs as a fixed sequence of stages and writes its results to disk. A verification stage re-checks every number. The dashboard is one HTML file. It opens in any browser and needs no server, no build step, and no network access.